

CE213 Linear Algebra: Introduction

Weeks: 1th – 2th

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Course Objectives

The aim of this lecture is to introduce the fundamental concepts of linear algebra and their applications in computer engineering.

Course Content:

Linear Equation Systems, Matrices, Determinants, Vector spaces, Linear Transformations, Matrix Representations of Linear Transformations, Inner Product Spaces, Eigenvalues and Eigenvectors.

Weekly Topics

Week	Topics
1. Week	Introduction, Linear Equation Systems (LES), Matrices
2. Week	Matrix Multiplication, Algebraic Properties of Matrix Operations, Special Types of Matrices
3. Week	Echelon Form of a Matrix, Solution of Linear Equation Systems
4. Week	Elementary Matrices, Finding the Inverse of a Matrix
5. Week	Determinants, Cramer's Rule
6. Week	Vector Spaces, Subspaces
7. Week	Spanning, Linear Independence
8. Week	Midterm
9. Week	Basis and Dimension
10. Week	Inner Product Spaces and Gram-Schmidt Method
11. Week	Linear Transformations, Kernel and Image of Linear Transformations
12. Week	Matrix Representation of Linear Transformations
13. Week	Eigenvalues and Eigenvectors
14. Week	Diagonalization

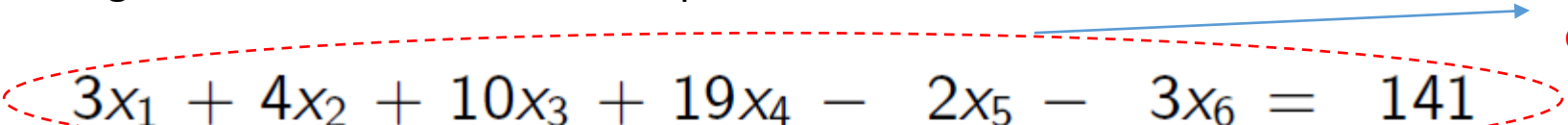
What is Linear Algebra?

- Linear:
 - Having to do with lines/planes/etc.
 - For example, $x + y + 3z = 7$, and **not not sin, cos ; not log; not x^2 , etc. Those are non-linear functions.**
- Algebra
 - Solving equations involving numbers and symbols,
 - from al-jabr (Arabic), meaning reunion of broken parts,
 - 9th century Abu Ja'far Muhammad ibn Muso al-Khwarizmi.

What is Linear Algebra?

Engineers need to solve lots of equations in lots of variables as shown below:

Linear
equation


$$3x_1 + 4x_2 + 10x_3 + 19x_4 - 2x_5 - 3x_6 = 141$$

$$7x_1 + 2x_2 - 13x_3 - 7x_4 + 21x_5 + 8x_6 = 2567$$

$$-x_1 + 9x_2 + \frac{3}{2}x_3 + x_4 + 14x_5 + 27x_6 = 26$$

$$\frac{1}{2}x_1 + 4x_2 + 10x_3 + 11x_4 + 2x_5 + x_6 = -15$$

This representation of the problem is enough to know some information about the set of solutions without having to solve the equations at all!

Large classes of engineering problems, no matter how huge, can be reduced to linear algebra:

$$A.x = b \text{ or}$$

$$A.x = \lambda x$$

What is Linear Algebra?

- Linear algebra is the study of vectors and linear functions.
- Vectors are things you can add and linear functions are functions of vectors that respect vector addition.

- A vector such as $v = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$, has 3 components $v_{11} = -1$, $v_{21} = 0$, $v_{31} = 1$

- A matrix is a rectangular array of numbers and/or variables. For instance

- $M = \begin{bmatrix} 1 & -1 & 2 & 0 \\ 1 + \pi & 0 & \pi & -1/3 \\ 3 & 1 & 2 & \sin(\pi/7) \end{bmatrix}$ is a matrix with 3-rows and 4-columns

as a notation $[M_{3 \times 4}]$. Some components of $[M_{3 \times 4}]$ are $M_{11} = 1$, $M_{23} = \pi$, $M_{34} = \sin(\pi/7)$. Matrice $[M_{3 \times 4}]$ has four-column vectors as indicated by red-lines.

Linear Equations

- A system of linear equations are sets of equations in the form of:

Diagram illustrating a system of linear equations (LES) in the form $A.X = b$.

The system consists of three linear equations:

$$\begin{cases} 2x_1 + 3x_2 + 5x_3 = 5 \\ 3x_1 + 1x_2 - 1x_3 = 2 \\ -2x_1 + 1x_2 + 1x_3 = 1 \end{cases}$$

Labels and arrows indicate the components:

- Coefficients (A):** Indicated by blue arrows pointing to the numerical values (2, 3, -2, 3, -2) in the equations.
- Variables (X_i):** Indicated by red arrows pointing to the variables (x_1, x_2, x_3) in the equations.
- Constant term (b):** Indicated by a green arrow pointing to the constant values (5, 2, 1) in the equations.

A linear equation is highlighted by a blue box around the first equation.

The system is summarized as $A.X = b$.

A system of linear equations (LES: Linear Equation Systems)

I believe you know how to solve it.



Linear Equations

- A system of linear equations:

m
equations
(rows)

$$\left\{ \begin{array}{l} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{array} \right.$$

n variables (columns)

m and n are integer numbers standing for **row count** and **column count**, respectively.

Linear Equations

- Solving a system of linear equations

$$\begin{array}{rcl} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n & = & b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n & = & b_2 \\ \vdots & & \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n & = & b_m \end{array}$$

Diagram illustrating a system of linear equations with variables $x_1, x_2, \dots, x_n$ and coefficients a_{ij} . The equations are represented by a matrix of coefficients a_{ij} and a vector b_i . The diagram includes three colored boxes with text: a yellow box labeled "span" pointing to the first equation, a blue box labeled "Independent" pointing to the second equation, and a green box labeled "rank" pointing to the third equation.

Does it have
solution?

Approximate
Solution

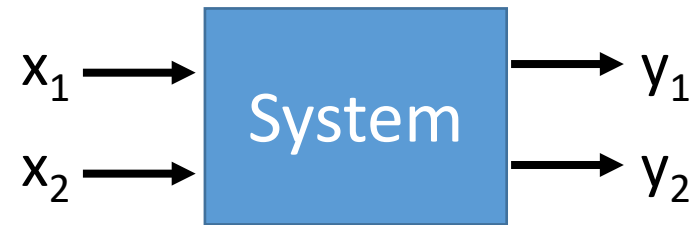
Does it have
unique solution?

How to find
the solution?

Determinants
($\square \square \square$)

Beyond 3 X 3

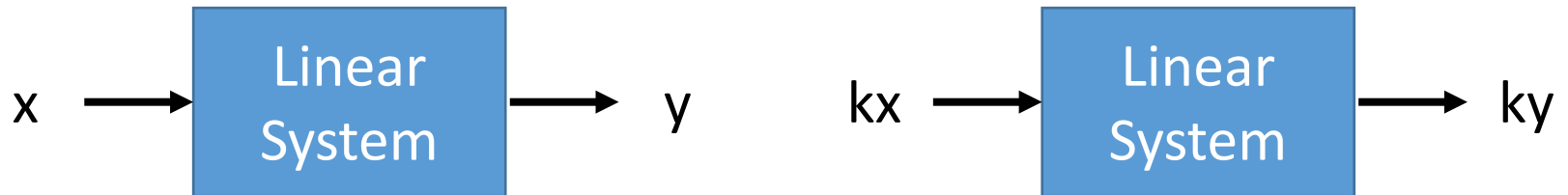
Linear System



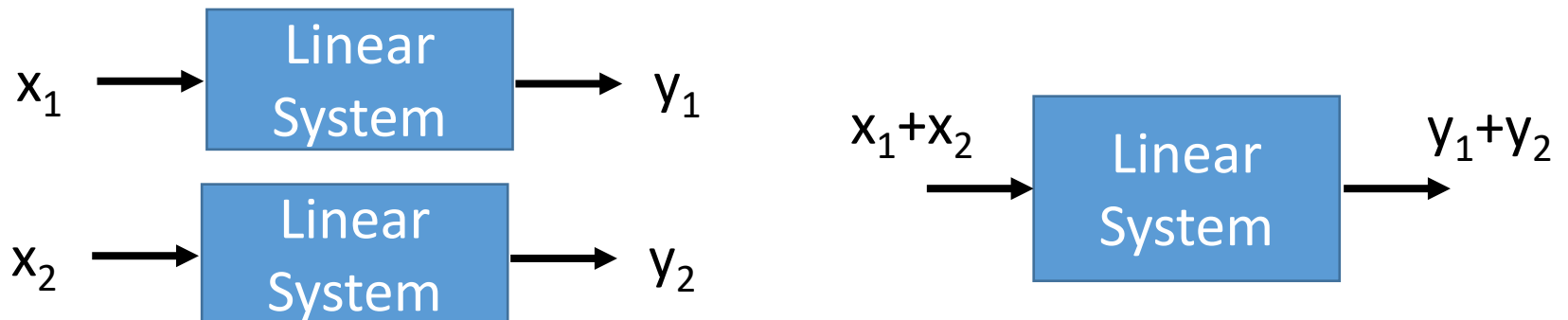
- We use a system of linear equations to describe a linear system
- What is system?
 - function, transformation, operator
 - A system has input and output
 - E.g. Speech Recognition System: The input is audio signals, and the output is recognition results.
 - E.g. Siri: You say “Hi, Siri”, it reply “What can I do for you?”
 - E.g. Communication system
 - Can have multiple inputs and outputs

Linear System

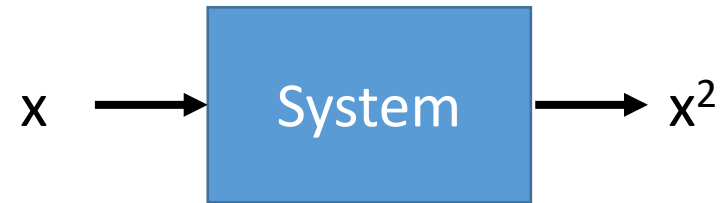
- Linear system have two properties
 - 1. Persevering Multiplication



- 2. Persevering Addition

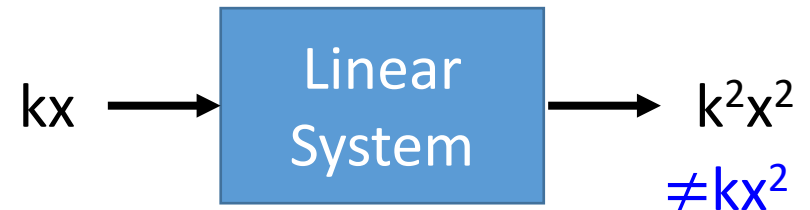
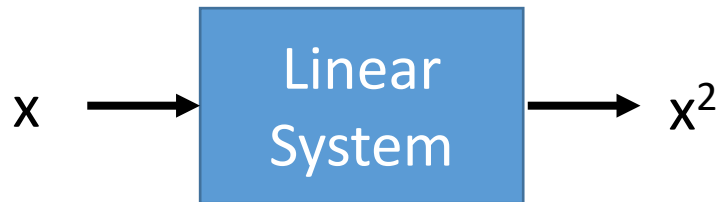


Linear System

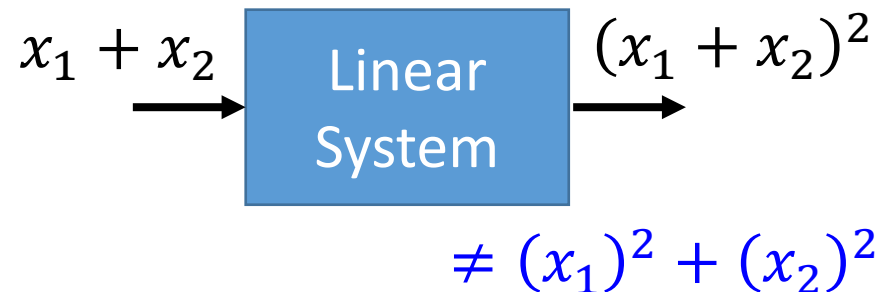
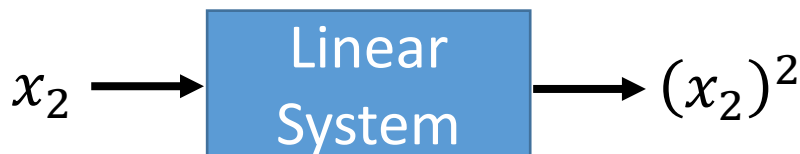
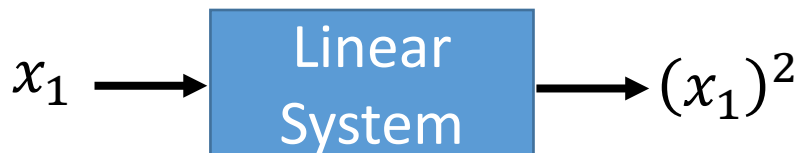


- Linear system have two properties

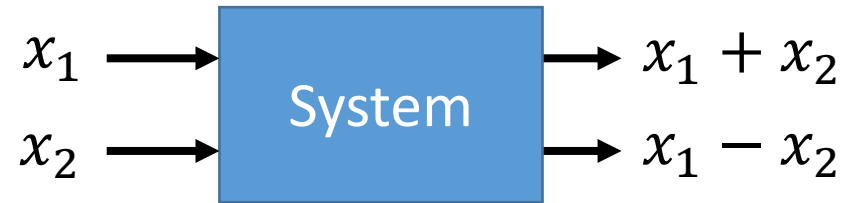
- 1. Persevering Multiplication



- 2. Persevering Addition

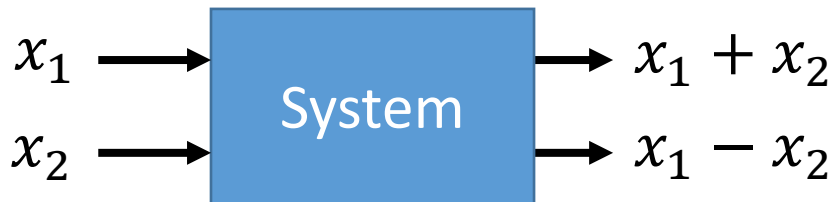


Linear System

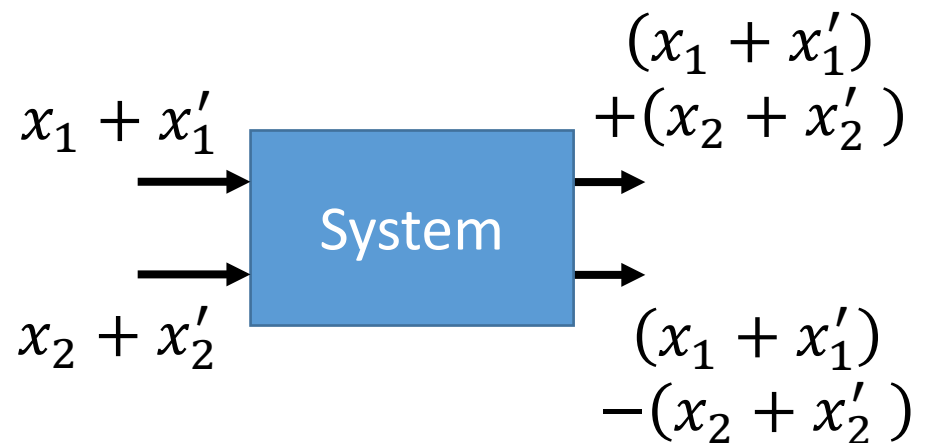
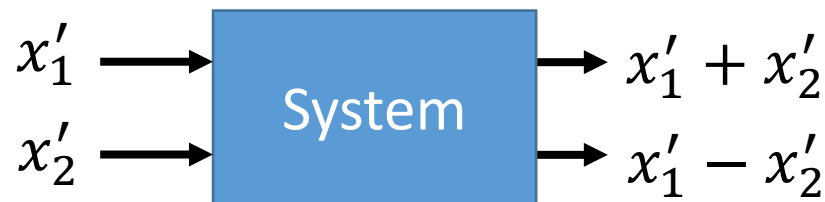
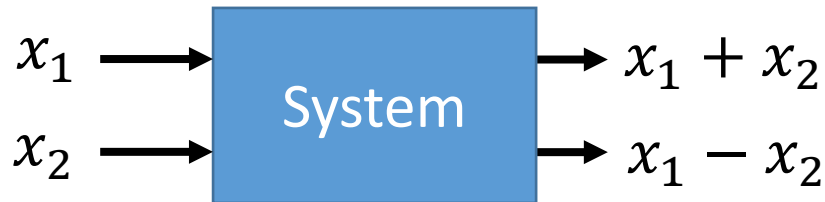


- Two properties

- 1. Persevering Multiplication

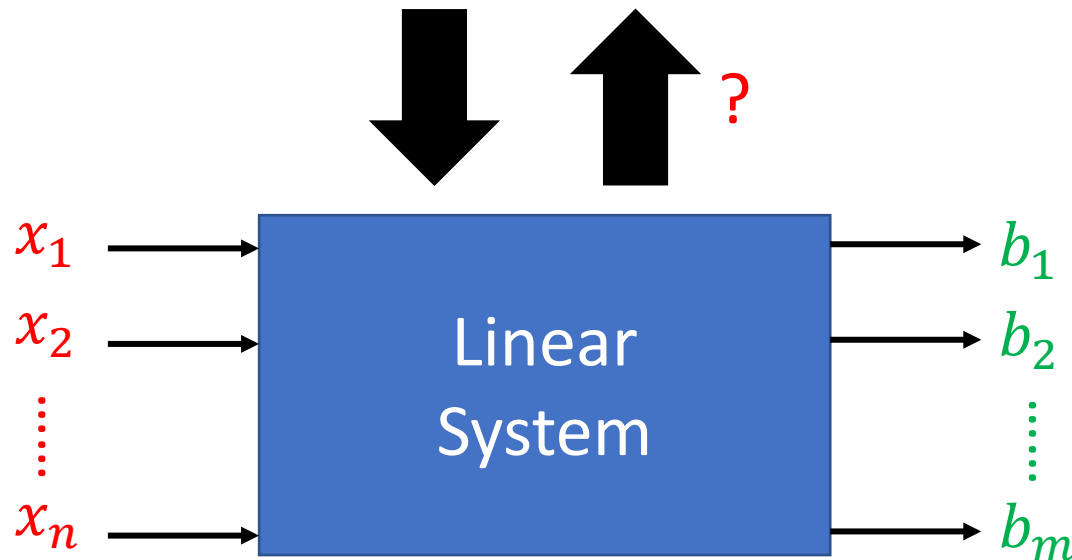


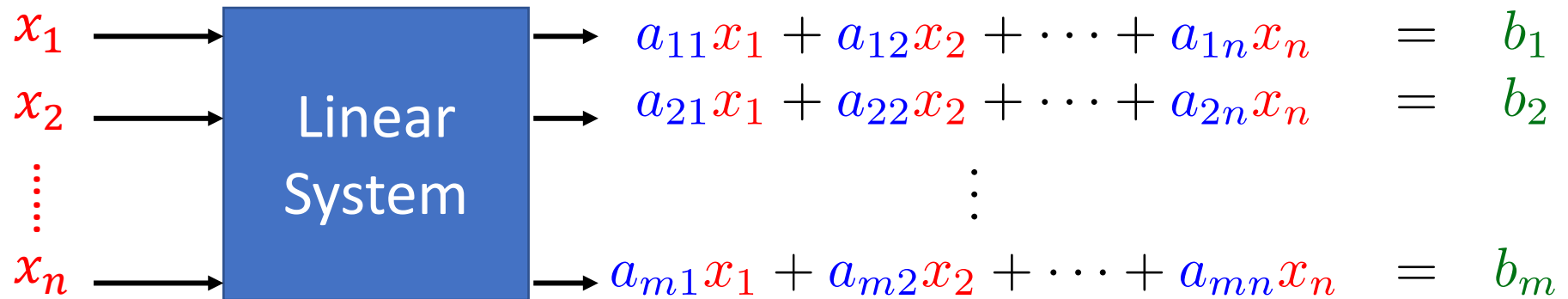
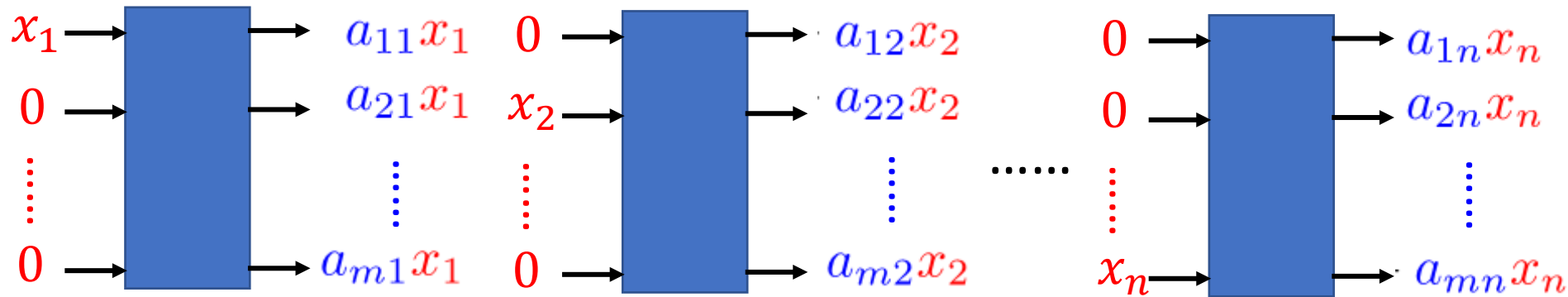
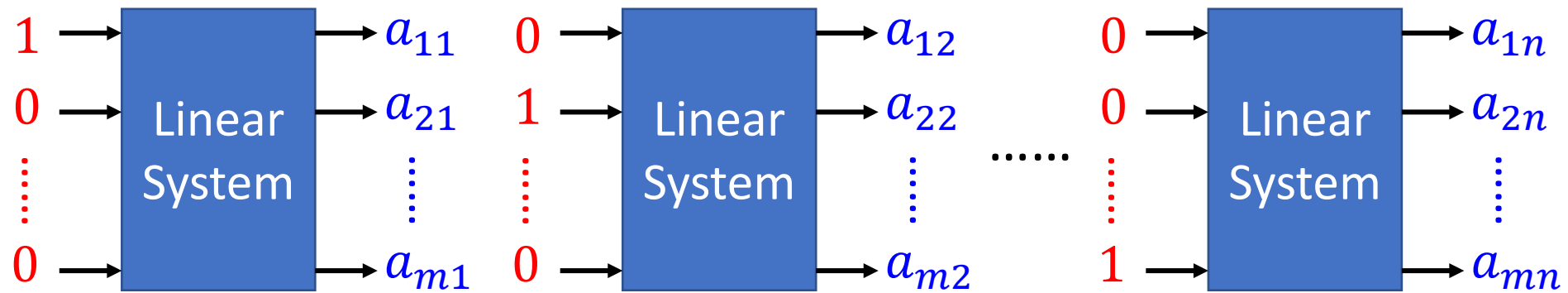
- 2. Persevering Addition



Linear System v.s. System of Linear Equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$



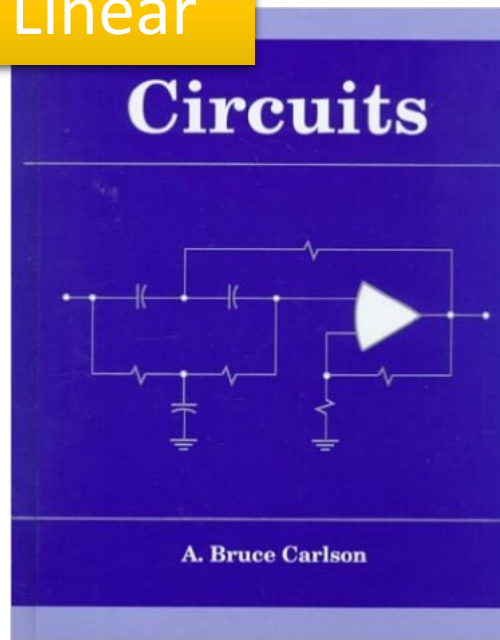


A linear system is described by a system of linear equations

Applications

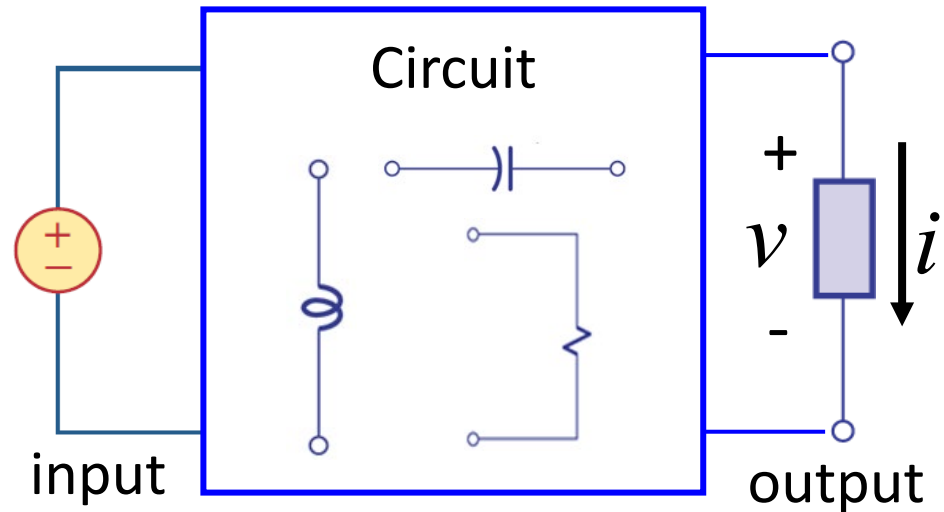
Most system are (assumed to) be linear

Linear



(□ □ □)

Input: voltage source, current source
output: voltage and current on the load.

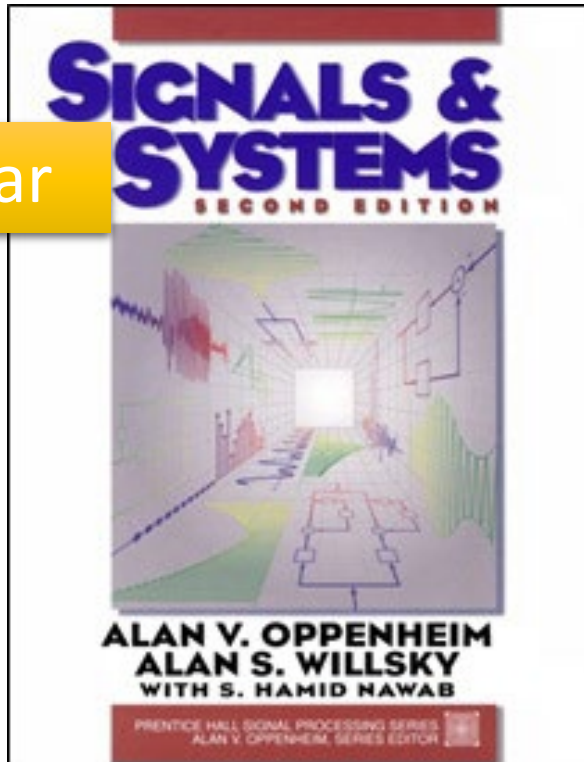


Linear System

Applications

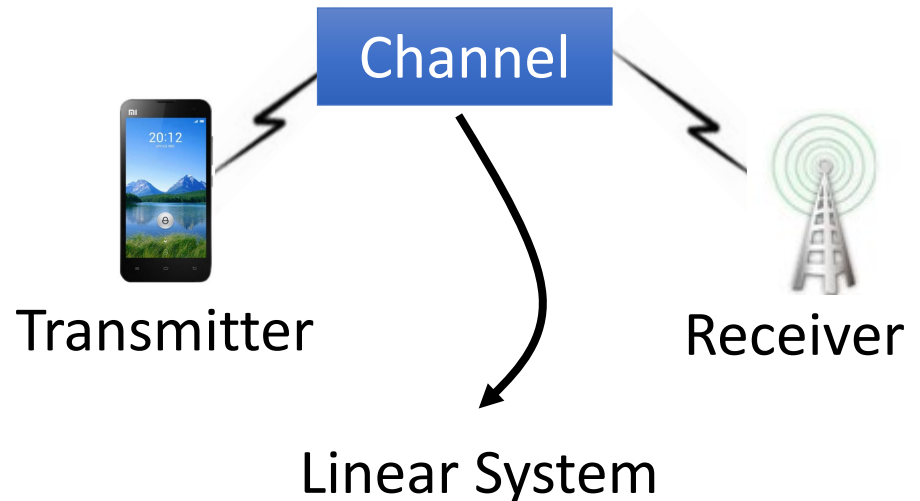
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Linear



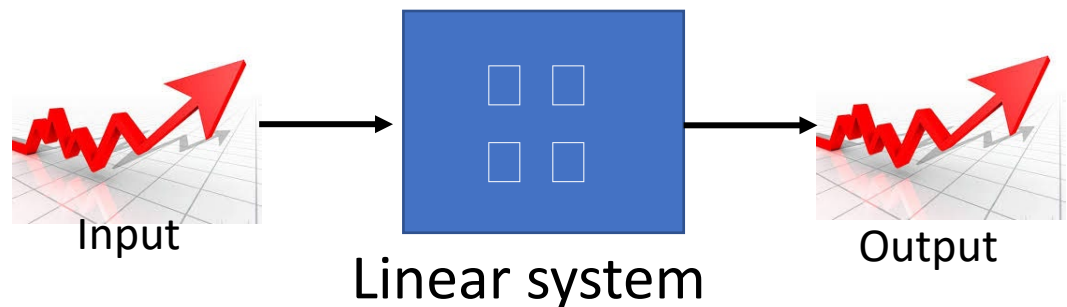
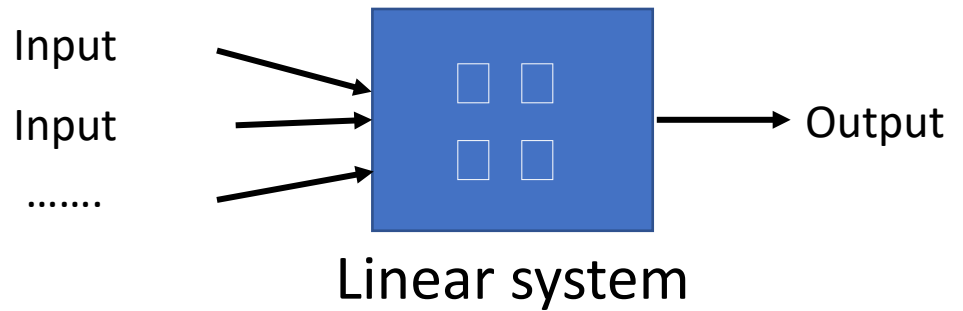
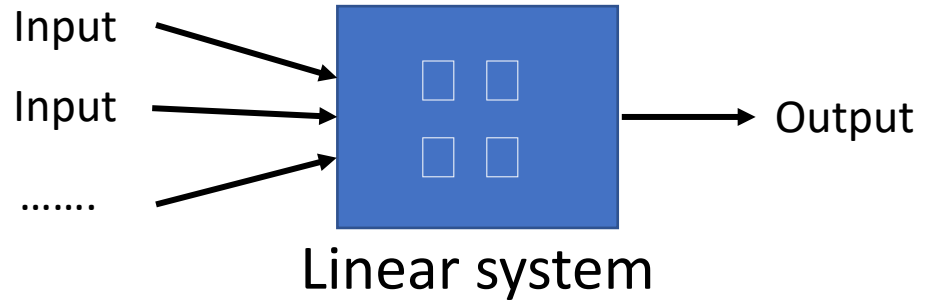
(□ □ □)

Communication System



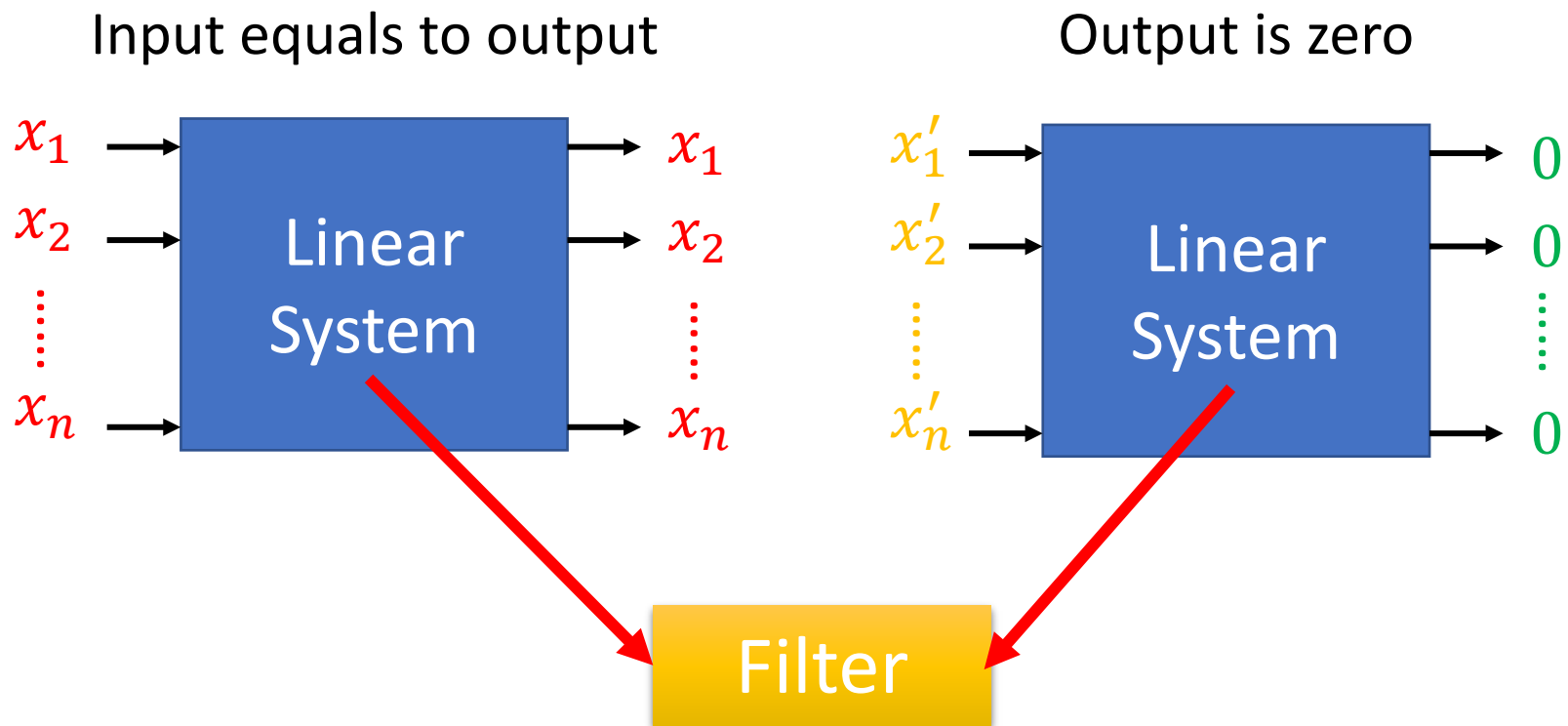
Applications

- Machine Learning
 - E.g. Prediction



Applications

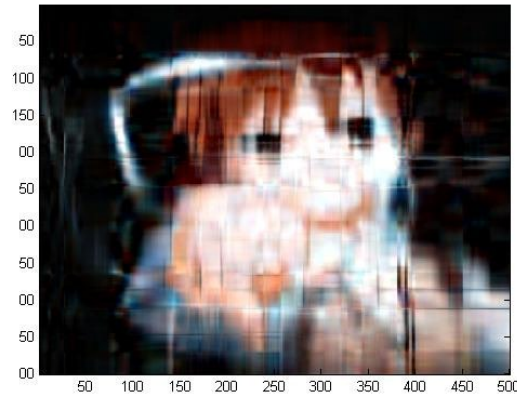
- We will learn EigenXXXXX



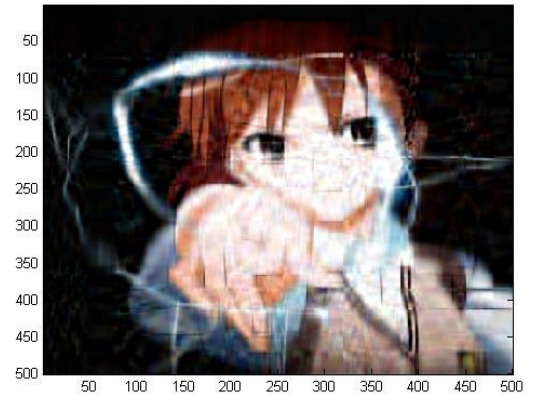


Applications – Image Compression

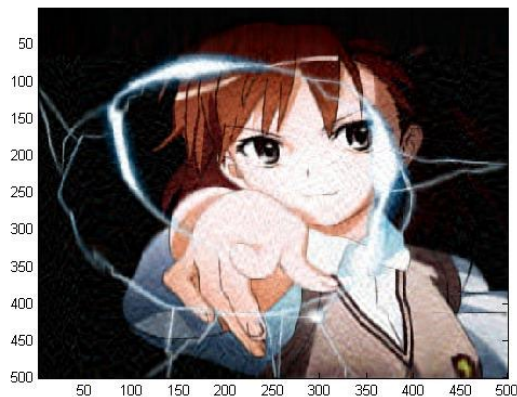
Noise: 25



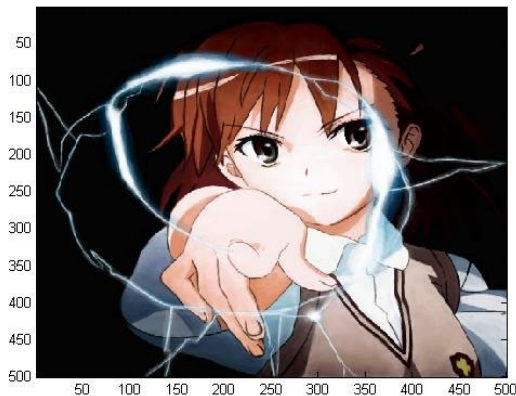
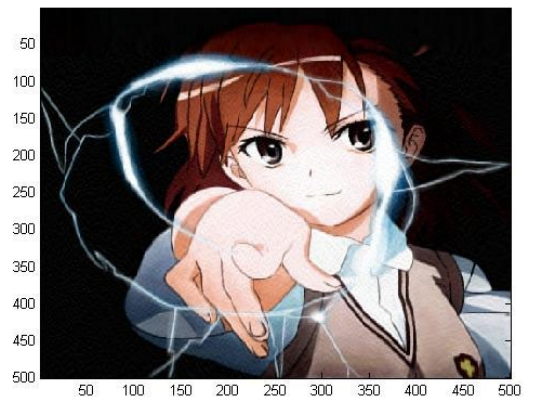
Noise : 12.5



Noise: 5



Noise: 2.5



Ground Truth

Linear Algebra is Important:

Computer Programming,
Image Processing,
Artificial Intelligence (AI),
System Modelling and Understanding, Forecasting,
Economy,
Energy,
...
...

Books:

Cemal Koç, "Basic Linear Algebra", METU Mathematics Foundation, 1996

B. Kolman, D. Hill, Elementary Linear Algebra with Applications, 9th edition, Pearson, 2008.

H. Anton, C. Rorres, Elementary Linear Algebra: Applications Version, Wiley; 11th edition, 2013.

Elementary Linear Algebra [12 ed.] 1119268044, 9781119268048, English Pages 611, 2019.

Elementary Linear Algebra, 11th Edition, by Howard Anton and Chris Rorres, 2014.

